THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8 light-minutes	Approx.
2	**(d)** 60 kg	Mass does not change.
3	**(d)** Earth's axis is tilted as it goes around the Sun	Tilt.
4	**(d)** waxing	Increasing.
5	**(b)** one sixth	Weaker gravity.
6	**(b)** stars appear to move across the night sky	Apparent motion.
7	**(d)** 30 March	About 29.5 days.
8	**(c)** 16	1440/90.
9	**(a)** halve	12 hours.
10	**(a)** sunset	Opposite the Sun.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	228/150 (1); about 1.5 cm (1).	1+1
12	They orbit the Sun (1); Closer than the stars (1).	1+1
13	Earth's rotation speed is highest (1); Gives extra boost (1).	1+1
14	24 hours = 1440 minutes (1); 1440/100 = 14.4 orbits (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	5000/10 (1); 500 days (1); Year/day (1).	1+1+1
16	Gravity pulls planets (1); Pull bends their path (1); Without it they would fly off in straight lines (1).	1+1+1
17	Moon is a quarter of the way around (1); Half of the lit side visible (1); Looks like a half disc (1).	1+1+1
	OR: 365/27 (1); About 13.5 (1); Approximately 13 complete orbits (1).	1+1+1
18	Long orbits (1); Take decades or centuries (1); Appear near the Sun (1).	1+1+1
	OR: Moon has no air or water (1); Little erosion (1); Earth's weather erases them (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Choose a scale for sizes (1); Place the Sun at the centre (1); Planets in order (1); Label features (1); Show orbits (1).	1+1+1+1+1
	OR: Moon's phase depends on position (1); Not on place (1); Rise times differ with longitude (1); Earth rotates (1); Same shape seen (1).	1+1+1+1+1
20	New moon starts the cycle (1); First quarter day 8 (1); Full moon day 15 (1); Last quarter day 22 (1); Cycle about 29.5 days (1).	1+1+1+1+1
	OR: GPS (1); Medical imaging (1); Water purifiers (1); Materials (1); Weather prediction (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 467	42000/90.
2	**(d)** Mercury and Venus	Inner planets.
3	**(b)** west	Near the setting Sun.
4	**(d)** Mangalyaan	Mars Orbiter Mission.
5	**(a)** Jupiter	A storm.
6	**(d)** Venus	Evening star.
7	**(b)** Chandrayaan-3	India's success.
8	**(d)** Uranus	Tilt about 98 degrees.
9	**(d)** its distance from the Sun	Farther means longer.
10	**(b)** weather forecasting by space	Satellite launched by rocket.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It faces away from the Earth (1); It is not lit (1).	1+1
12	Cost and knowledge (1); Safety and cooperation (1).	1+1
13	600/6 (1); 100 N (1).	1+1
14	Gravity is weaker (1); Mass stays the same (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	5 minutes = 300 s (1); 300 x 3 lakh (1); 9 crore km (1).	1+1+1
16	Far from the Sun (1); Less sunlight (1); Thick gases (1).	1+1+1
17	Collisions with satellites (1); Damage to space stations (1); Cost of replacement (1).	1+1+1
	OR: Sun is 150 million km away (1); Proxima is about 4 light years (1); Sun is nearer (1).	1+1+1
18	Scale the distances (1); Mark positions on a rope (1); Compare the spacing (1).	1+1+1
	OR: Communications (1); Remote sensing (1); Disaster warnings (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Bright city lights (1); Hide faint stars (1); Shielded lamps (1); Switch off unnecessary lights (1); Dark-sky areas (1).	1+1+1+1+1
	OR: Inner small and rocky (1); Outer large and gaseous (1); Outer have rings (1); Few moons for inner (1); Many for outer (1).	1+1+1+1+1
20	Backup for humanity (1); Scientific research (1); No breathable air (1); Cold and radiation (1); Costly (1).	1+1+1+1+1
	OR: Used for direction (1); Calendar and seasons (1); Saptarishi (1); Observation records (1); Aryabhata used observations (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Beyond Earth** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Jupiter, Earth, Mars, Mercury	Correct.
2	**(a)** rotate faster	Rotation period.
3	**(a)** atmosphere blurs the view	Air and light pollution.
4	**(b)** the Moon's history	Geology.
5	**(c)** Pole star lies almost along its extension	So the Pole star appears fixed.
6	**(c)** clear view without air disturbance	Better images.
7	**(d)** it can damage satellites	Collision risk.
8	**(c)** the Sun's bright light hides it	It remains.
9	**(a)** radio waves take time to travel the distance	Finite speed.
10	**(d)** solar panels	Solar cells.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Thick carbon dioxide atmosphere (1); Greenhouse effect (1).	1+1
12	Earth rotates and has a tilt (1); The Moon rotates slowly (1).	1+1
13	Atmospheric motion (1); Planets are discs (1).	1+1
14	It lies near the northern horizon (1); Earth's shape and latitude (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Eclipse: shadow of Earth or Moon (1); Phase: change in the lit part visible (1); Eclipse is occasional (1).	1+1+1
16	It moves fast sideways (1); Falls around the Earth (1); Orbit (1).	1+1+1
17	An optical illusion (1); Comparison with objects (1); Real size is the same (1).	1+1+1
	OR: Lower gravity (1); No air resistance (1); Less fuel needed (1).	1+1+1
18	From the light and colour (1); From size and brightness (1); Compared with models (1).	1+1+1
	OR: 75,000,000/25,000 (1); 3000 hours (1); About 125 days (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Early sky watching (1); Telescope (1); Space probes (1); Rovers (1); Space telescopes (1).	1+1+1+1+1
	OR: Photosynthesis (1); Water cycle (1); Solar panels on rooftops (1); Satellites use panels (1); Clean energy (1).	1+1+1+1+1
20	Low orbits for imaging (1); High geostationary for TV (1); Medium for navigation (1); Different coverage (1); Different speeds (1).	1+1+1+1+1
	OR: Knowledge (1); Technology (1); Disaster management (1); Cost issues (1); Balance with needs (1).	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.